ABDULKADIR EROL

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EDUCATION

University of Central Florida, Orlando, USA *August 2025 - Present*
Ph.D. in Computer Science (Expected Spring 2030)
Advisor: Prof. Cüneyt Gürcan Akçora

Georgia State University, Atlanta, USA *August 2023 - June 2025*
M.Sc. in Computer Science
Advisor: Prof. Uğur Kursuncu

EPITA, Paris, France *February - May 2022*
Erasmus+ Exchange Program in Data Science & Analytics

Bilkent University, Ankara, Turkey *September 2017 - July 2023*
B.Sc. in Computer Engineering

RESEARCH INTERESTS

Large Language Models (LLMs), Topological Data Analysis (TDA), Parameter-Efficient Fine-Tuning (PEFT), Continual Learning, Machine Unlearning

PUBLICATIONS

1. **TopoTuner: Topological Finetuning of Large Language Models**
   A. Erol, Y. Mahajan, V. Hariprashad, B. Rababah, S. Karmaker, C. G. Akcora, M. Shah
   Submitted PAPER
2. **TopoLearner: Topological Continual Learning for Large Language Models**
   A. Erol, T. G. B. Ngo, S. Karmaker, C. G. Akcora, M. Shah
   Submitted
3. **Echoes of Human Malice in Agents: Benchmarking LLMs for Multi-Turn Online Harassment Attacks**
   T. Padhi, P. Lu, **A. Erol**, T. Sutar, G. Sharma, M. Sonmez, M. De Choudhury, U. Kursuncu
   AAAI ICWSM, 2027 PAPER
4. **Unmasking Toxicity and Vulnerabilities in Large Vision-Language Models**
   A. Erol, T. Padhi, A. Saha, M. E. Aktas, U. Kursuncu
   ACM TIST, 2026 PAPER
5. **From Reddit to Generative AI: Evaluating Large Language Models for Anxiety Support Fine-tuned on Social Media Data**
   U. Kursuncu, T. Padhi, G. Sinha, **A. Erol**, J. K. Mandivarapu, C. R. Larrison
   Submitted PAPER

RESEARCH EXPERIENCE

Graduate Research / Teaching Assistant (GRA/GTA) *August 2025 - Present*
University of Central Florida, Orlando, FL

- **Topological Data Analysis:** Applied persistent homology to analyze topological changes in LLM attention matrices and identify projections for selective freezing and updating.
- **PEFT & Continual Learning:** Developed selective freezing strategies for efficient fine-tuning and selective updating strategies to mitigate catastrophic forgetting across sequential tasks.
- **Model Merging & Machine Unlearning:** Exploring topology-guided model merging and modularity, with applications to machine unlearning and selective removal of learned capabilities.
- **Teaching Assistant:** *Blockchains and Smart Distributed Contracts* (CIS 5730, Summer 2026) and *Artificial Intelligence for FinTech* (CAP 5619, Fall 2026).

Graduate Research Assistant (GRA) *August 2023 - June 2025*
Georgia State University, Atlanta, GA

- **Toxicity Analysis & Red Teaming:** Explored vulnerabilities in LVLMs, including *LLaVA, InstructBLIP, Fuyu, Qwen, DeepSeek, Gemini, GPT, and Grok*, using persuasive prompting strategies informed by social theories; applied *hard/soft prompts, prompt engineering, and fine-tuning.*
- **LLMs in Mental Health:** Conducted comprehensive evaluations of *Llama and GPT* for mental health applications, focusing on sympathy, empathy, bias, toxicity, readability, and coherence.

SELECTED COURSES

UCF: Computer Vision, Computer Vision Systems, Advanced Computer Vision, AI Knowledge Systems, Current Topics in Machine Learning, Interactive Data Visualization
GSU: Machine Learning, Advanced Machine Learning, Introduction to Deep Learning, Deep Learning, Advanced Text Processing & LLMs, Data Mining, Advanced Bioinformatics
Bilkent: Discrete & Combinatorial Mathematics, Probability & Statistics for Engineers, Introduction to Statistics, Linear Algebra & Differential Equations, Algorithms

SKILLS

Programming and Libraries: Python, NumPy, scikit-learn, PyTorch, TensorFlow, Keras
LLMs and Fine-Tuning: Hugging Face Transformers, PEFT, LoRA, DeepSpeed
Topological Data Analysis: Ripser, Persim, Persistent Homology, Wasserstein Distances
Data Visualization: Matplotlib, Seaborn, Plotly
Graph Neural Networks (GNNs): PyTorch Geometric, NetworkX
NLP: NLTK, spaCy

WORK EXPERIENCE

TUBITAK YTE, Ankara, Turkey — *Candidate Engineer* *February - August 2023*

- Developed a Product Tracking System for the Turkish Medicines and Medical Devices Agency using Java (Spring) and PostgreSQL in a 29-member team.

Azerion, Istanbul, Turkey — *Summer Intern* *August - September 2022*

- Built real-time peer-to-peer chat and video conferencing using Socket.io, WebRTC, Firebase, and Postman; developed a Twitch-like platform donation system using JavaScript and React.

Netcad Software, Ankara, Turkey — *Summer Intern* *June - July 2021*

- Developed an autonomous sustainability system using JavaScript, HTML, and CSS.

EXTRACURRICULAR ACTIVITIES

- **IEEE Bilkent Student Branch**. Publicity Coordinator (2018-2019), IAS Founding Chairman (2019), Vol – E Project Founding Coordinator (2019-2020), Executive Committee Treasurer (2020-2021), Counsellor (2022-2023).
- **Social Awareness Projects**. Intermediate in Turkish Sign Language. Organized events for *children with hearing loss* and taught sign language to more than 100 college students (2017-2022).
- **IEEE Turkey Student Branches** Publicity & Design YouTube Coordinator (2019-2020).

AWARDS AND ACHIEVEMENTS

- Bilkent University Comprehensive Scholarship: Full tuition waiver & stipend during the B.Sc.
- Ranked **934th/2M** in university and **18th/150k** in postgraduate entrance exams in Turkey
- **Valedictorian** in Yozgat Şehitler Science High School, GPA: 97.6/100
- National Chess Tournament Finalist in Turkey